

Earth's Coordinates

Guided Notes — ANSWER KEY

Driving question: How do we locate things on Earth?

1. Why we need a coordinate system

Directions given by landmark (“two miles past the old barn”) only work if you already know the landmark. A **coordinate system** gives every point on Earth one unique address built from a grid of intersecting lines.

The grid is anchored to something that never moves: Earth's axis of rotation. The poles and the equator come from the way Earth spins, not from a human decision.

2. Latitude and longitude

Complete the table.

	Latitude	Longitude
Lines run...	<u>east–west</u>	<u>north–south</u>
Measured...	<u>north–south</u>	<u>east–west</u>
Do they touch?	<u>no, parallel</u>	<u>yes, at the poles</u>
Measured from...	<u>the equator</u>	<u>prime meridian</u>
That line is...	0° latitude	0° longitude
Range	0° to <u>90°</u> N or S	0° to <u>180°</u> E or W

Every coordinate needs **a number AND a direction**. 43° is not a location. 43° N is a line; 43° N, 76° W is a point.

Latitude is always written first.

Why does latitude stop at 90° but longitude go to 180°?

At 90° you have reached the pole — there is no more Earth to travel. Longitude can run 180° each way because you can circle halfway around in either direction.

3. Degrees and minutes

When we zoom in, latitude and longitude are given in degrees *and* **minutes**, written with a prime symbol (').

$$1^{\circ} = \underline{\quad 60 \quad}' \quad 30' = \frac{1}{2}^{\circ} \quad 10' = \frac{1}{6}^{\circ}$$

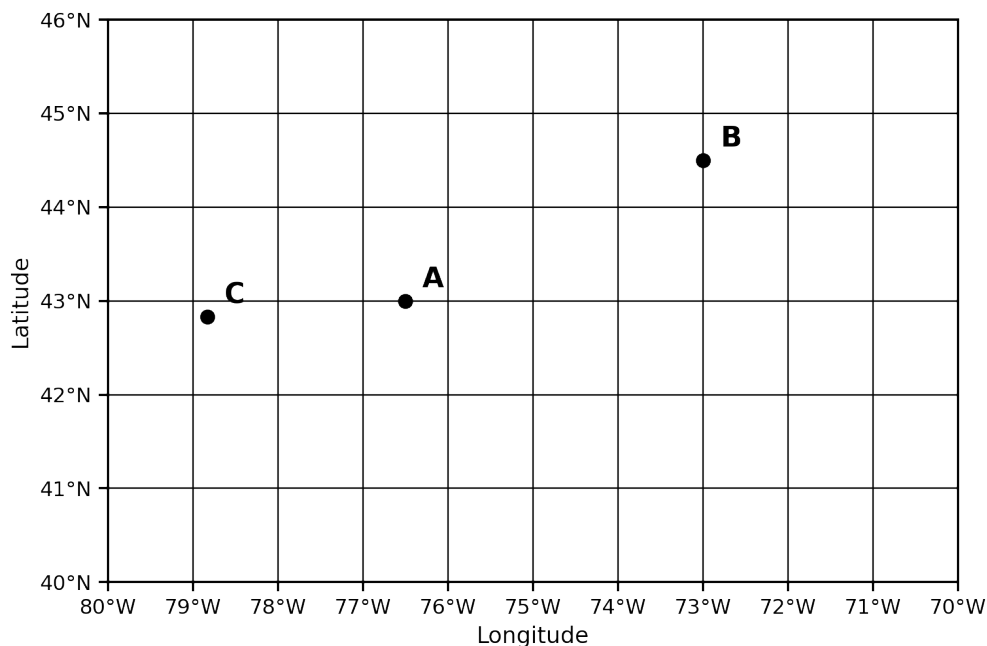
- Halfway between two degree lines is about 30'.
- We estimate minutes to the nearest 10 minutes.

Watch out: 76.5° is **not** $76^{\circ}50'$. Half a degree is 30', just like half an hour is 30 minutes.

Worked example. Syracuse, NY sits on the 43° parallel and a little past the 76° meridian: **43° N, $76^{\circ}10'$ W**

4. Practice: reading the grid

Give the latitude and longitude of each point to the nearest 10 minutes. Remember the hemisphere letters.



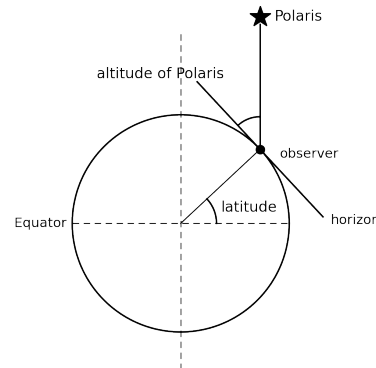
	Latitude	Longitude
A	<u>43°00' N</u>	<u>76°30' W</u>
B	<u>44°30' N</u>	<u>73°00' W</u>
C	<u>42°50' N</u>	<u>78°50' W</u>

5. Latitude and Polaris

Polaris — the **North Star** — sits almost directly above Earth's axis of rotation.

Because of that, the **altitude** of Polaris (its angle above the horizon) is equal to the observer's latitude.

Polaris can only be seen from the Northern Hemisphere.



Check the extremes. Fill in the altitude of Polaris at each location.

Where you are	Latitude	Altitude of Polaris
North Pole	90° N	<u>90°, overhead</u>
Albany, NY	42° N	<u>42°</u>
Equator	0°	<u>0°, on horizon</u>
Southern Hemisphere	—	<u>not visible</u>

Moving around. Circle or fill in what happens to the altitude of Polaris.

- Travel **north**: Polaris gets higher and latitude increases.
- Travel **south**: Polaris gets lower and latitude decreases.
- Travel **east or west**: Polaris does not change and latitude stays the same.

Moving 1° of latitude changes the altitude of Polaris by exactly 1° — about 111 km (69 miles) of travel on the ground.

6. Longitude and time zones

Meridians are counted east and west of the prime meridian until they meet at 180°, which is called the Int'l Date Line.

$$\frac{360^\circ \text{ of rotation}}{24 \text{ hours}} = \underline{15}^\circ \text{ per hour}$$

Time zones are 15° of longitude wide, and each one represents 1 hour of time.

Earth rotates from west to east, so places to the **east** rotate into sunlight first and their clocks are ahead. Places to the **west** have clocks that are behind.

Worked example. Albany is 74° W. Denver is 105° W. How many hours apart are their clocks, and which is earlier?

$$105^\circ - 74^\circ = \underline{\mathbf{31^\circ}} \quad \frac{31^\circ}{15^\circ/\text{h}} \approx \underline{\mathbf{2 \text{ hours}}}$$

Denver is *west* of Albany, so Denver's clock is **2 hours behind**.

7. Check for understanding

1. The angle from the horizon to Polaris is equal to the _____ of the observer. [1]
A. latitude
B. longitude
2. Meridians are 15° apart, so time zones are separated by [1]
A. 4 hours
B. 24 hours
C. 1 hour
3. A ship sails due east across the Atlantic Ocean. The altitude of Polaris will [1]
A. increase
B. decrease
C. stay the same
4. How many minutes are in 1° of latitude? [1]
A. 10
B. 60
C. 100

Exit ticket

1. Using the *Generalized Bedrock Geology of New York State* map in your ESRT, give the latitude and longitude of Albany, NY to the nearest 10 minutes. [2]

Solution: Approximately $42^\circ 40'$ N, $73^\circ 45'$ W.

2. An observer measures the altitude of Polaris as 30° . What is their latitude, and which hemisphere are they in? Explain how you know the hemisphere. [2]

Solution: They are at 30° N, in the Northern Hemisphere. Polaris cannot be seen at all from the Southern Hemisphere, so any observer who can measure its altitude must be north of the equator.

3. Explain in one sentence why time zones are 15° of longitude wide.

[2]

Solution: Earth rotates 360° in 24 hours, which works out to 15° per hour, so each 15° of longitude corresponds to one hour of time.

Vocabulary

Latitude	angular distance north or south of the equator, 0° to 90°
Longitude	angular distance east or west of the prime meridian, 0° to 180°
Equator	the 0° line of latitude, halfway between the poles
Prime meridian	the 0° line of longitude, through Greenwich, England
Parallel	another name for a line of latitude
Meridian	another name for a line of longitude
Altitude	the angle of an object above the horizon
Int'l Date Line	the 180° meridian, where the calendar date changes